

# Automatic Handwriting Quality Analysis

## Data Strategy for Overcoming inconsistencies in Ground Truth

### 1. Introduction

#### 1.1. Motivation

Handwriting is...

- a key cultural technique of humanity, key for cultural development and knowledge,
- vital for the development of certain motor and intellectual skills in children, and
- an important factor in learning for both children and adults.

For assessment, supporting initial learning and the improvement of writing,

- an objective measure for the “quality” of handwritten documents is needed.
- Manual scales exist but are laborious and prone to subjectivity.

#### 1.2. Current Problems of Automatic Approaches

1. **Feature selection is inconsistent** and often not clearly motivated.

→ We don't know, what does what and what matters

2. Widespread **non-interpretability**, which is undesirable.

→ **Feedback on handwriting is not actionable**

3. Ground truth (what defines quality) is acquired very differently

→ **Comparison of approaches is difficult**

Research Question:

*Which approaches to automatic handwriting quality analysis classify most accurately, and does feature interpretability constrain accuracy?*

## 2. Comparing Approaches

Multiple datasets of handwritten documents are combined for >2500 samples total.

1. Approaches will be **categorized** (expecting ~3–4 categories)
2. For each category, a **representative is chosen and implemented**
3. **Representatives are evaluated** against a test-portion of the dataset

### 2.1. Acquiring Ground Truth Labels

- **Automatic Labelling** by aggregation of
  1. multiple approaches with available source-code, or
  2. multiple LLMs / classification models.
- The **automatic labelling is first validated** on a subset of the data (50–100 samples) **against two human readers**.
- Human readers grade with **standardized scales assessing the most important criteria** from literature (Table 1).
- When correlation is good, all >2500 samples can be automatically labelled
  - **Notion of quality is grounded in literature**
  - **Ground Truth is reproducible**

| Factor     |                          | Description                                                           |
|------------|--------------------------|-----------------------------------------------------------------------|
| Legibility | Global Legibility        | Whether the document is legible on the first read.                    |
|            | Letter Recognizability   |                                                                       |
|            | Size                     | Uniformity and adequacy of letter and sentence size.                  |
|            | Letter Shape             | Whether letters are normatively well-formed.                          |
|            | Letters and word spacing | Uniformity of letter and word spacing.                                |
|            | Line adherence           | Uniformity of letter placement within lines of text.                  |
|            | Layout                   | Uniformity of document margins.                                       |
|            | Effort to read           | How easy a document feels to read.                                    |
|            | Alterations              | Amount of alterations such as strikeouts, re-tracing of letters, etc. |
|            | Colour of line           |                                                                       |
|            | Slant                    | Uniformity and angle of slant.                                        |
| Speed      |                          | Time to write a reference text.                                       |
| Fluency    |                          | Changes in speed over time.                                           |

Table 1: Prominent factors impacting general handwriting quality from an initial exploratory literature review.